Name: _____

Midterm Exam #1 for Math 104, Fall 2015

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
8	10	
9	10	
10	10	
11	10	
12	10	
Total	120	

- Please show **ALL** your work on this exam paper. No credit will be given without work.
- **NO** books, class notes, laptops, cell phones, calculators, or any other electronic devices may be used during the exam.
- You **ARE** allowed one 8.5 x 11 " sheet of notes.
- No form of cheating will be tolerated. You are expected to uphold the Code of Academic Integrity.

My signature below certifies that I have complied with the University of Pennsylvania's Code of Academic Integrity in completing this midterm examination.

Name (printed): _____

Signature: _____

Date: _____

1. Evaluate the definite integral $\int_0^{\pi/2} e^{-x} \sin x \, dx$.

   (a) $1 - e^{-\pi/2}$

   (b) $1 + e^{-\pi/2}$

   (c) $\frac{1}{2} + \frac{e^{-\pi/2}}{2}$

   (d) $\frac{1}{2} - \frac{e^{-\pi/2}}{2}$

   (e) 0

2. Calculate $\int_0^{\pi/2} \sin^3 \theta \cos^2 \theta \, d\theta$

   (a) 1

   (b) $2/15$

   (c) $3\pi/5$

   (d) $\pi/2$

   (e) $1/5$

3. $\int_0^{\pi/4} \sin(5x)\sin(7x)\,dx$

   (a) 1/2
   (b) 1/4
   (c) $\pi/35$
   (d) $\pi/2$
   (e) 1/3

4. $\int_0^{\pi/4} \tan^3(\alpha)\sec^4(\alpha)\,d\alpha$

   (a) 5/12
   (b) 1/2
   (c) 1/3
   (d) 5/6
   (e) 7/8

5. $\int \frac{\sqrt{9-x^2}}{x^2} dx$

6. $\int \frac{1}{e^x \sqrt{e^{2x}+3}} dx$

7. Write down the integral that computes the volume of the solid of revolution obtained by revolving the region bounded by the graphs of $f(x) = (x-2)^2$ and $g(x) = x$ from $x = 1$ to 4 around the y-axis. You do not have to integrate it.

8. Find the arc length of the function $f(x) = (4 - x^{2/3})^{3/2}$ between $x = 0$ and $x = 2$.

   (a) $3(2)^{2/3}$

   (b) $3(2)^{3/2}$

   (c) $3(3)^{2/3}$

   (d) $2(3)^{3/2}$

   (e) $2(3)^{3/2}$

9. Find the voluming of the solid generated by revolving the area bounded by $y = x^2$ and $y = 2x$ around the line $y = -1$.

   (a) $\frac{51\pi}{5}$
   (b) $\frac{51\pi}{7}$
   (c) $\frac{4\pi}{15}$
   (d) $\frac{104\pi}{15}$
   (e) $\frac{52\pi}{15}$

10. Use the shell method to find the volume of the solid generated by revolving the region bounded by the curve $y = -(x-1)^2 + 1$ and the x-axis about the y-axis.

11. Find the surface area of the surface generated by revolving $f(x) = \sqrt{x}$ for $0 \leq x \leq 4$ about the x-axis.

(a) $\frac{1}{6}(17\sqrt{17})\pi$

(b) $\frac{1}{6}(17\sqrt{17} - 17)\pi$

(c) $(17\sqrt{17} - 1)\pi$

(d) $\frac{1}{6}(17\sqrt{17} - 1)\pi$

(e) $\frac{5}{6}(17\sqrt{17} - 1)\pi$

12. Find the center of mass of a thin plate covering the region bounded by the graph of $y = \cos(2x)$, $y = 0$ and $x = \pm\pi/4$ if the plates density at the point (x, y) is $\delta(x) = 2$.

    (a) $(2, \pi/4)$

    (b) $(2, \pi/8)$

    (c) $(0, \pi/8)$

    (d) $(0, \pi/6)$

    (e) $(0, \pi/4)$

13. Determine the limit of the sequence $a_n = \frac{2n-3^n}{6^n}$ if it exists.

14. Determine the limit of the sequence $a_n = n \ln\left(1 + \frac{3}{n}\right)$ if it exists.

15. Find the indefinite integral

$$\int \frac{x^3 + x^2 - x - 2}{x^3 - x}\, dx\,.$$

16. Determine if the following improper integral converges

$$\int_2^\infty \frac{1}{x^3 - x + 1} dx$$

17. The lifetime of a particular brand of lightbulb is modeled by the pdf

$$p(x) = \frac{1}{10000} e^{-\frac{1}{10000}x}$$

What is the probability that a lightbulb will last between 2000 and 5000 hours.

18. Let $\int_{-\infty}^{\infty} \frac{C}{1+x^2}\, dx$ and state what value of C makes this into a probability density. What is the mean of a random variable X with this probability density function?

19. Determine whether

$$\sum_{n=1}^{\infty} \frac{2n}{n^2+1}$$

converges or diverges

20. Define a sequence by $a_1 = 27$ and

$$a_n = \frac{3}{2 + a_{n-1}}$$

You may assume the limit $\lim_{n\to\infty} a_n$ exists. Calculate it.

21. Calculate

$$\int_0^1 \frac{1}{\sqrt{x}} dx$$

22. What is the sum of the series

$$\sum_{k=2}^{\infty} \frac{3}{2^k}$$

23. What is the sum of the series

$$\sum_{k=0}^{\infty} \frac{2}{k^2 + 3k + 2}$$

is it converges.

24. Does

$$\sum_{k=1}^{\infty} \frac{1}{1 + n^2}$$

converge? What is an upper and lower bound for the series.